Gases & Absolute Temperature

Exam code: 0625 & 0972

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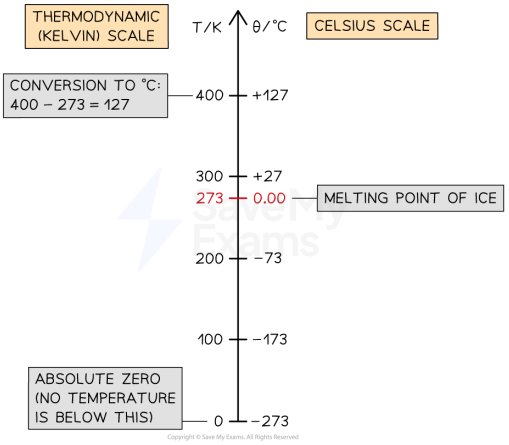
Absolute temperature

What is absolute temperature?

- Temperature measured in kelvin is called **absolute temperature**
- The **kelvin temperature scale** begins at **absolute zero**
  - **0 K is equal to –273 °C**
  - An increase of 1 K is the same change as an increase of 1 °C
- It is not possible to have a temperature **lower** than 0 K
- This means a temperature in kelvin will **never** have a negative value
- To convert between temperatures θ in the Celsius scale, and T in the Kelvin scale, use the following conversion:

$$\theta / {}^{\circ}C = T / K - 273$$

$$T / K = \theta / {}^{\circ}C + 273$$



Conversion chart relating the temperature on the Kelvin and Celsius scales

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

and Celsius scales of temperature.

a) 0 K = _____ °C

b) 0 °C = _____ K

c) 20 °C = _____ K

Part (a)

Step 1: Choose whether to add or subtract 273 to the value

- The question is in kelvin therefore subtract 273 to convert to Celsius

Step 2: Do the calculation

- $0 - 273 = -273$

Step 3: Write the answer with units

- 0K = −273 °C

Part (b)

Step 1: Choose whether to add or subtract 273 to the value

- The question is in Celsius therefore add 273 to convert to kelvin

Step 2: Do the calculation

- $0 + 273 = 273$

Step 3: Write the answer with units

- 0 °C = 273 K

Part (c)

Step 1: Choose whether to add or subtract 273 to the value

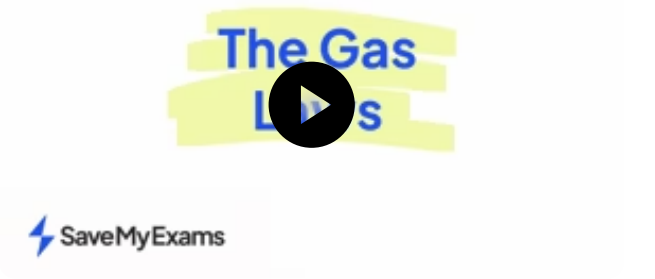
- The question is in Celsius therefore add 273 to convert to kelvin

Step 2: Do the calculation

- $20 + 273 = 293$

Step 3: Write the answer with units

- 20 °C = 293 K

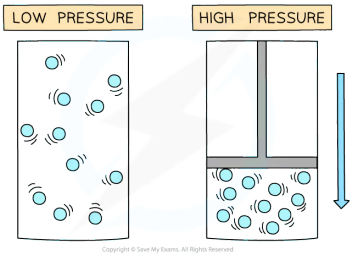


Did this video help you? Yes No

The gas laws

Pressure & volume (constant temperature)

- If the temperature of a gas remains **constant**, the pressure of the gas changes when it is:
  - **Compressed** – decreases the volume which **increases** the pressure
  - **Expanded** – increases the volume which **decreases** the pressure



At constant temperature, changing the volume changes the pressure

- Similarly, a change in pressure can cause a change in volume
- A **vacuum pump** can be used to remove the air from a sealed container
- The diagram below shows the change in volume to a tied up balloon when the pressure of the air around it decreases:
  - The balloon is tied, so there are a **fixed number** of air particles within it
  - At **normal pressure**, the air pressure outside the balloon is **greater** than the air pressure inside the balloon
  - When **air is removed** from the bell jar, the air pressure outside the balloon is **less than** the air pressure inside the balloon
  - The **space between** the air particles in the balloon increases, so the **volume** of the balloon increases

At constant temperature, changing the pressure changes the volume

- When a gas is compressed, the molecules will hit the walls of the container **more frequently**
  - This creates a larger overall **net force** on the walls, which increases the **pressure**

Pressure & temperature (constant volume)

- **Increasing temperature increases the pressure** of a gas which is kept at a **constant volume**
- The **average speed** of molecules increases when the temperature increases (and vice versa)
- As the gas heats up, the molecules will travel at a higher speed
  - They collide with the walls more often and with greater force, increasing the pressure

- Therefore, at a constant volume, an **increase** in temperature **increases** the pressure of a gas and vice versa
  - Diagram A shows molecules in the same volume collide with the walls of the container more as the temperature increases
  - Diagram B shows that since the temperature is directly proportional to the pressure (at constant volume), the graph is a straight line

Did this video help you? Yes No

Boyle's Law

Extended tier only

Boyle's law

- If the temperature *T* of an ideal gas is constant, then **Boyle's Law** is given by:

$$p \propto \frac{1}{V}$$

- This means the pressure is **inversely proportional** to the volume of a gas
- This can also be written as:

$$pV = \text{constant}$$

- The relationship between the pressure and volume for a fixed mass of gas at constant temperature can also be written as:

$$p_1 V_1 = p_2 V_2$$

- Where:
  - p_1 = initial pressure (Pa)
  - p_2 = final pressure (Pa)
  - V_1 = initial volume (m^3)
  - V_2 = final volume (m^3)

- Notice that volume and pressure are measured in m^3 and Pa respectively
  - In calculations if units are given in cm^3 or MPa this is a rare case where calculations can be done using the original units **as long as answers are reported in the same, original units and the final and initial units match**

Boyle's Law graph: Pressure is inversely proportional to volume

Examiner Tips and Tricks

It is an easy mistake to make to think that an **inversely proportional** graph will be a straight line sloping downwards. After all, a directly proportional graph is a straight line (through the origin) which slopes upwards!

The curve above which 'tends towards zero' (meaning the curve gets closer and closer but never touches the axis, or zero is an inversely proportional curve, as the graph below shows.

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